

Vehicle A/C Diagnostic Enclosure V4

Orion ANSYS static-structural screening report - public STEP release

SCREENING ONLY - NOT A SAFETY CERTIFICATION

The run uses an idealized 250 N compression load and a nominal isotropic PETG material card. It does not model layer anisotropy, creep, temperature, vibration, fasteners, gasket contact, impact, water ingress, or vehicle service loads.

Executive readout

All three ANSYS MAPDL solve points completed. Mesh-integrated volume matched the imported CAD volume within 0.00075%. Maximum displacement was nearly stable between the 4.0 mm and 2.5 mm meshes, but peak von Mises stress increased materially at the finest mesh. The analysis is useful for identifying deformation scale and stress-sensitive regions; it is not converged enough, nor physically calibrated enough, to support a strength or service-life claim.

9.71 MPa	0.820 mm	70,228	0.0007%
Finest-mesh peak stress	Finest-mesh displacement	SOLID187 elements	Maximum volume error

Provenance

Source model	V4 body STEP from the public design-evolution archive
Source SHA-256	e84119d4285ede12fa56ab8cf4b2404c2a76e70c07955359e06313563581c844
Orion job	mdl_60ee65aca9294544
Tool chain	FreeCAD 1.0 -> Gmsh -> ANSYS MAPDL 2022 R2
Repository	https://github.com/IsaiahDupree/3d-print-library
Run completed	2026-09-20 23:22 EDT (2026-09-21 03:22 UTC)

Method and boundary conditions

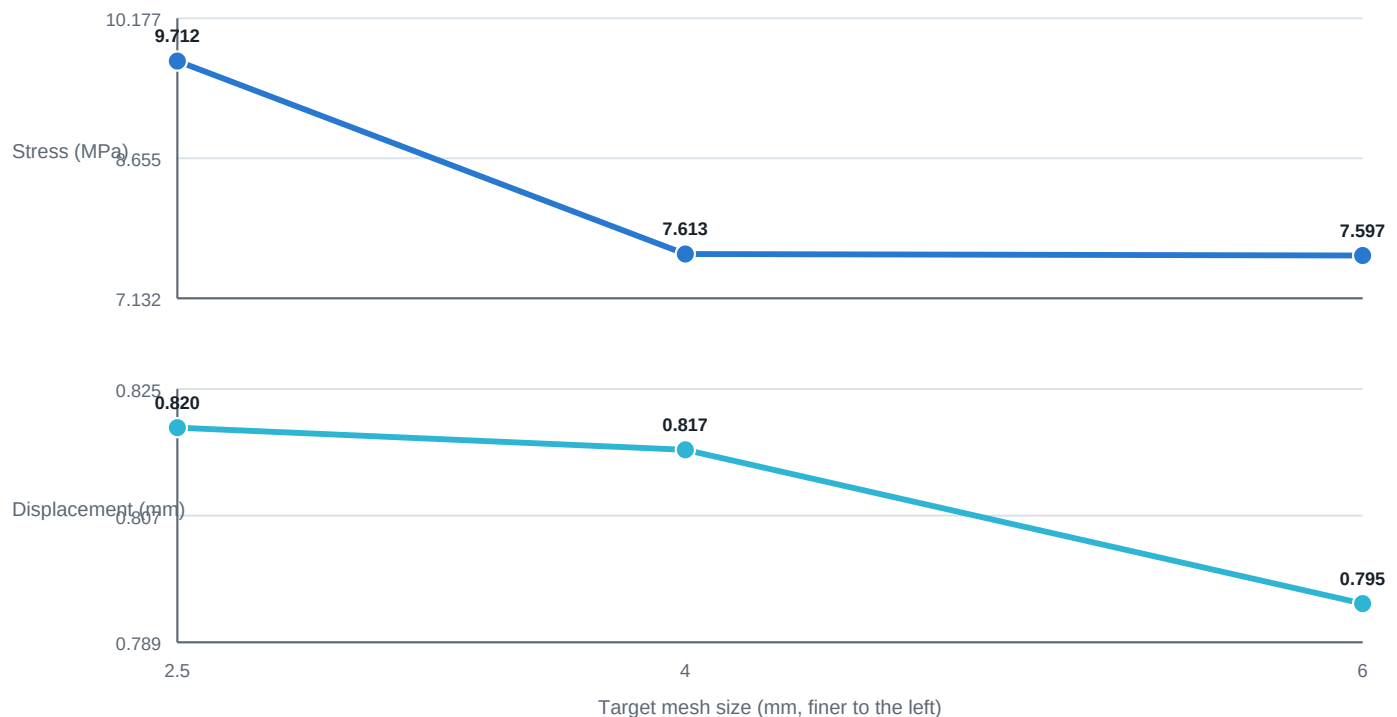
The uploaded STEP was checked as one valid solid, translated through OpenCASCADE/Gmsh, and meshed with 10-node SOLID187 tetrahedra. The model was normalized to its bounding-box minimum. Nodes on the minimum-Z face were fixed in all degrees of freedom; a total -Z force of 250 N was distributed across nodes at the maximum-Z lid-interface face.

Geometry	159.8 x 60.0 x 88.0 mm; CAD volume 160,990.53 mm ³ ; 195 faces
Fixture region	tub_base_support: normalized bounding-box minimum Z, +/-0.01 mm
Load region	lid_interface_top: normalized bounding-box maximum Z, +/-0.01 mm
Load	250 N total distributed force in -Z
Material	Nominal isotropic PETG screen: E = 2330 MPa, nu = 0.40, density = 1.27 g/cm ³
Mesh sweep	6.0 mm, 4.0 mm, 2.5 mm target sizes
Outputs	Peak averaged von Mises stress, maximum displacement, mass, mesh/CAD volume agreement

Mesh sweep results

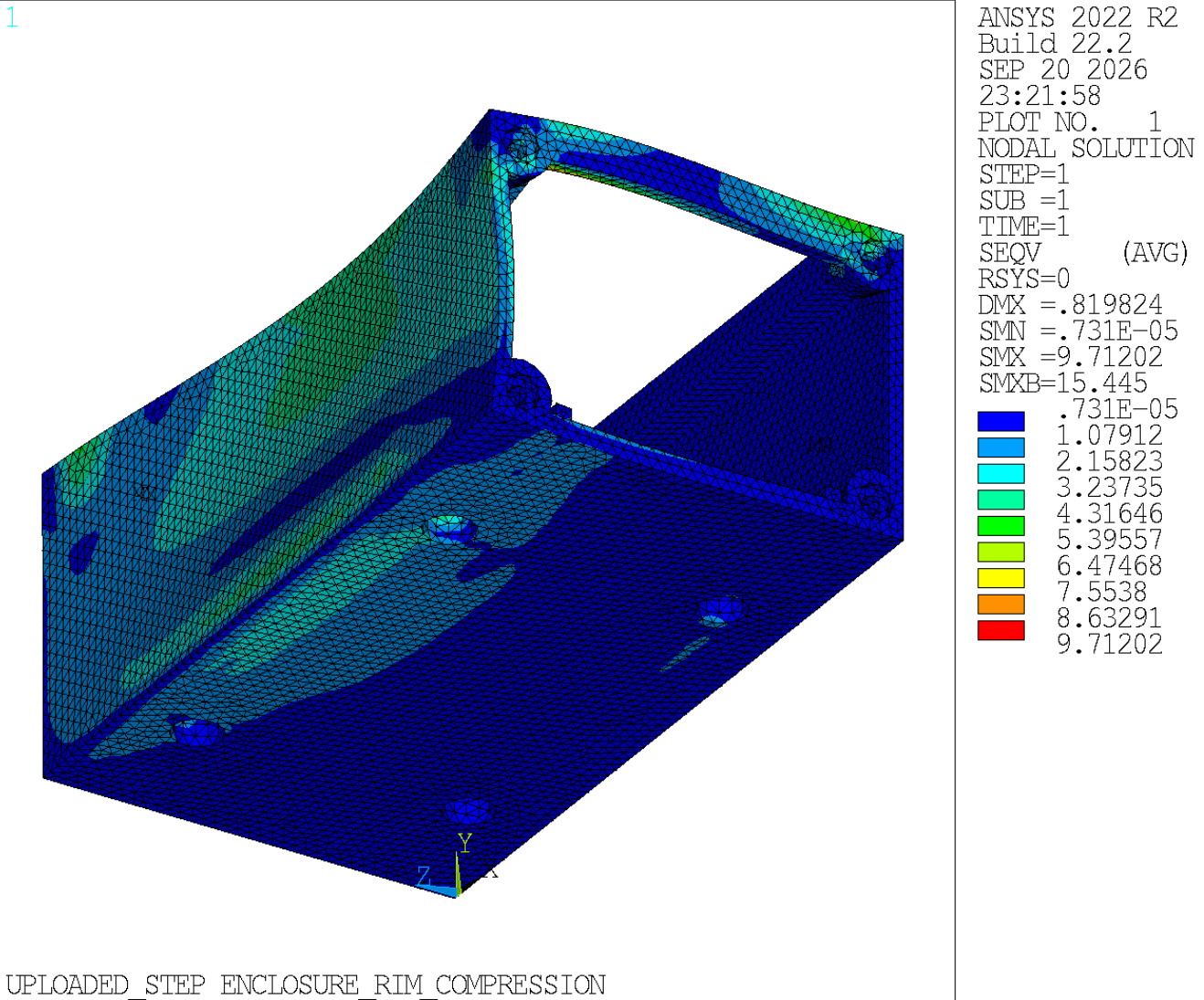
Mesh	Nodes	Elements	Fixture / load nodes	Stress	Displacement	Volume error
6.0 mm	32,186	16,397	1,371 / 1,395	7.597 MPa	0.7949 mm	0.000745%
4.0 mm	55,136	28,239	2,919 / 2,919	7.613 MPa	0.8167 mm	0.000475%
2.5 mm	132,710	70,228	7,365 / 7,357	9.712 MPa	0.8198 mm	0.000465%

Mesh convergence - 250 N idealized rim compression



Displacement changed 0.38% from the 4.0 mm to 2.5 mm mesh; peak stress changed 21.6%. This indicates displacement is approaching stability while local peak stress is still mesh-sensitive.

Finest-mesh stress field



ANSYS MAPDL averaged von Mises stress contour for the 2.5 mm target mesh. The image is a diagnostic location map, not a pass/fail plot. Peak stress is 9.712 MPa and remains mesh-sensitive; the nominal material card intentionally supplies no allowable or printed-part strength claim.

Engineering interpretation

What this run supports	What this run does not support
- Exact digital provenance to the public STEP hash. - Three successful solver points with low volume-transfer error. - Approximate displacement scale under one idealized compression case. - Identification of regions that deserve finer local study or geometry changes.	- No factor of safety, yield, fatigue, creep, impact, or service-life claim. - No printed-layer anisotropy, infill, perimeter, seam, temperature, or moisture model. - No fastener preload, gasket contact, vibration, vehicle mount, or crash/load-path qualification. - No approval for under-hood installation or driving use.

Required next gates before an engineering claim

1. Print material/process coupons on the intended machine, nozzle, orientation, perimeter count, infill, drying state, and temperature profile; measure tensile/compressive properties and creep at the intended temperature.
2. Replace the bounding-box support/load idealization with the actual mount, screw, gasket, and contact interfaces, then run local mesh refinement until peak stress is demonstrably stable.
3. Bench-test a printed empty enclosure under measured clamp loads, torque, thermal cycling, spray/leak exposure, and vibration. Record deformation and failure modes.
4. Complete the existing physical vehicle fit-gauge program. Do not place printed polymer in the vehicle bolt clamp path and do not carry refrigerant pressure in any printed part.
5. Independently review electrical protection, cable strain relief, pressure hardware, under-hood material suitability, and service procedures.

RELEASE VERDICT

Publishable as a transparent simulation-screened prototype report. Not qualified for vehicle installation, structural duty, pressure containment, or safety-critical service.

Artifact integrity

Artifact	SHA-256
Source public V4 body STEP	e84119d4285ede12fa56ab8cf4b2404c2a76e70c07955359e06313563581c844
Orion results.csv	07c15e604ef2f8b6826ff2afa5c1814cef00050a7d6adca7e346e9402ded070a
Orion manifest.json	10fd2292848180bc283d5227480c172cb7180125dddbc98ca5ed9f2602f37bd0
Orion sweep.png	ac8fef117e7d700ec4a7237d8ae7ce194f87c54c865ea9549c5f42d2188c5829
Finest contour_02.png	fd95eee1e3f4e47f21d1a8f6255df195cd7bc8a125253c1c7873e019ec499acd

Prepared for public release by Isaiah Dupree. Reproduce against the exact source hash and Orion job record above. Raw ANSYS solver outputs are retained in the private engineering evidence archive; the public package includes this report, the machine-readable manifest, results table, convergence chart, and contour images.